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ORIGINAL PAPER

Rapid determination of heavy metals and arsenic in herbal medicines using energy-dispersive X-ray fluorescence spectrometry

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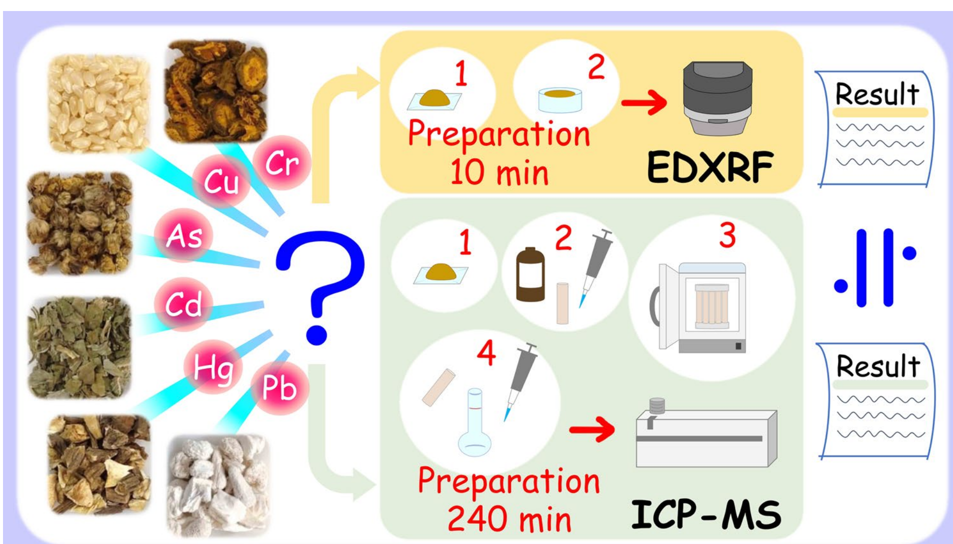
Abstract

The Heavy Metals Limit Test in the Japanese Pharmacopoeia is a chemical method used to detect the total amount of heavy metals based on Pb contents. However, determining the amounts of particular metals has become popular since the ICH-Q3D Guideline for Elemental Impurities was adopted. This is effectively performed using energy-dispersive X-ray fluorescence (EDXRF) analysis, an analytical method used to detect elements. In this study, we determined six elements (Cr, Cu, As, Cd, Hg, and Pb) in 25 herbal medicines using EDXRF and compared the quantitative values to those obtained using inductively coupled plasma-mass spectrometry (ICP-MS). The 25 herbal medicines used included 20 plants, two animals, and three minerals. EDXRF quantitative values for Cr, Cu, As, and Pb were obtained through calibration curve methods, whereas Cd and Hg were all below the limit of quantitation (10σ). Comparison of the quantitative values obtained via EDXRF and ICP-MS revealed ratios of 65–198%. The results for both methods were similar, with slightly larger differences observed for minerals between the EDXRF and ICP-MS quantitative values than those observed for plants. Some differences in quantitative values were observed in the comparison of six herbal medicines among different collection sites, owing to the environments in which Coptis rhizome and Mentha herb were grown. Furthermore, Longgu revealed significant elemental variations among the samples during simultaneous multi-elemental analysis. EDXRF enabled the easy and convenient determination of heavy metals in herbal medicines. Thus, it has potential applications in the quality control and evaluation of herbal medicines.

Graphical abstract



Keywords Heavy metal · Quantitative analysis · Simultaneous multi-elemental analysis · Herbal medicine · EDXRF · ICP-MS

Extended author information available on the last page of the article

Introduction

Heavy metals are abundant in natural environments. Although some heavy metals are important nutrients for plants and animals, others are highly toxic and should consequently be avoided. Therefore, controlling heavy metal content is extremely important for the safe distribution of herbal medicines. In this manuscript, herbal medicines indicate not only plant origins but also animal and mineral origins.

In the Japanese Pharmacopoeia (JP), the heavy metal limits for herbal medicines are specified mainly using roots and rhizomes via the Heavy Metals Limit Test 〈*1.07*〉 [1]. This limit test detects the total amount of heavy metals reacting with sodium sulfide TS in acidic solutions as a function of Pb content. However, the adoption of ICH-Q3D, the Guideline for Elemental Impurities [2], has demonstrated the need to measure specific metals. This trend is also observed for herbal medicines. ISO 18664:2015 [3], established by ISO/TC 249 (Traditional medicine), specifies the determination methods for each heavy metal in herbal medicines used in Traditional Chinese Medicine. The Chinese Pharmacopoeia [4] and United States Pharmacopeia Herbal Medicines Compendium [5] specify content limits for each heavy metal in some monographs. Furthermore, the Hong Kong Traditional Chinese Medicines Standards [6] uniformly specify content limits for each heavy metal in herbal medicines.

Methods for determining heavy metals include atomic absorption spectrophotometry, inductively coupled plasma-atomic emission spectrometry, and inductively coupled plasma-mass spectrometry (ICP-MS). These are listed according to the apparatus used in the Physical Methods in JP General Tests. In contrast, the Heavy Metals Limit Test specified for herbal medicines is listed in the Chemical Methods in JP General Tests and not instrumental analysis. Methods such as ICP-MS enable highly sensitive analysis of heavy metals. However, the installation and operation of the apparatus are costly.

Energy-dispersive X-ray fluorescence (EDXRF) is used for the elemental analysis of minerals, soils, and foods [7–9]. EDXRF is an analytical instrument for elementary types and contents; it measures the element-specific fluorescent X-rays generated from samples. This feature is markedly easy to operate and convenient, as it requires no preparations, such as the digestion and ashing of samples using high-concentration acid for analysis. Moreover, it is listed in the general test method of the United States Pharmacopeia—National Formulary and European Pharmacopoeia [10, 11].

In this study, we aimed to examine heavy metals in herbal medicines using EDXRF and evaluate the validity of EDXRF compared to ICP-MS. Our findings provide a basis for the potential application of EDXRF to ensure the safe assessment of heavy metals in herbal medicines.

Materials and methods

Materials

Twenty-five herbal medicines (20 plants, two animals, and three minerals) were selected from items listed in the JP, which included different types of medicinal parts or Materials. In total, 77 samples (Specimen Nos. NIHS-DPP-60001-60077) were used in this study, including 72 domestically distributed products purchased from 11 Manufacturers and 5 research samples. The details are presented in Table 1.

Reagents

For the EDXRF analysis, standard solutions of Cr (100 mg/l), Cu (1000 mg/l), and Pb (100 mg/l) were obtained from Kanto Chemical Co., Inc. (Tokyo, Japan). XSTC-2046, a mixed standard solution containing As, Cd, and Hg, was obtained from SPEX CertiPrep (Metuchen, NJ, USA). For the ICP-MS analysis, standard solutions of Cr (1000 mg/l), Cu (1000 mg/l), As (1000 mg/l), Cd (100 mg/l), Hg (100 mg/l), Pb (1000 mg/l), and Au (1000 mg/l) were obtained from FUJIFILM Wako Pure Chemical Corporation (Osaka, Japan). Nitric acid (electronic grade, purity 70%) was obtained from Kanto Chemical Co., Inc. Ultrapure water was purified using the Milli-Q Integral 5 system (Merck, Darmstadt, Germany).

Apparatus and measurement conditions

A Multi-beads Shocker™ (MB3200(S); Yasui Kikai Corporation, Osaka, Japan) was used to grind the samples into powder. An EDX-7200 fluorescence spectrometer (Shimadzu Corporation, Kyoto, Japan) was used for the EDXRF analysis, and PCEDX-Pro software (Ver. 2.07) was used for quantification. The measurement conditions are listed in Table 2. A microwave digestion system (ETHOS UP; Milestone, Italy) was used for pretreatment in the ICP-MS analysis. An Agilent 8800 ICP-QQQ spectrometer (Agilent Technologies Japan, Ltd., Tokyo, Japan) was used for the ICP-MS analysis, and the MassHunter 4.3 software was used for quantitation. The plasma mode was set to Low Matrix, the pump rotation speed to 0.1 rps, and the number of measurements to 1 point/peak.

Preparation of analytical samples

Seventy-seven samples (Table 1) were ground into powder using a polycarbonate tube and an agate ball mill.

EDXRF analytical samples: Sample powder (2.0 g) was weighed in a sample container with a 5-μm thick

Table 1 Samples for this study

Sample ID	Sample name	Collection sites	Collection year	Specimen No.
01-01	Phellodendron bark (黄柏)	Nagano, Japan	2020	NIHS-DPP-60001
02-01	Coptis rhizome (黄連)	Sichuan (四川), China	2018	NIHS-DPP-60002
02-02		Chongqing (重慶), China	2019	NIHS-DPP-60003
02-03		Chongqing (重慶), China	2009	NIHS-DPP-60004
02-04		Sichuan (四川), China	2018	NIHS-DPP-60005
02-05		Sichuan (四川), China	2016	NIHS-DPP-60006
02-06		Sichuan (四川), China	Unknown	NIHS-DPP-60007
02-07		Sichuan (四川), China	Unknown	NIHS-DPP-60008
02-08		Sichuan (四川), China	2020	NIHS-DPP-60009
02-09		Japan	Unknown	NIHS-DPP-60010
02-10		Japan	Unknown	NIHS-DPP-60011
02-11		Tamba region, Japan	Unknown	NIHS-DPP-60012
02-12		Fukui, Japan	2013, 2014	NIHS-DPP-60013
03-01	Chrysanthemum flower (菊花)	Henan (河南), China	2018	NIHS-DPP-60014
04-01	Brown rice (粳米)	Niigata, Japan	2012, 2021, 2022	NIHS-DPP-60015
04-02		Unknown	Unknown	NIHS-DPP-60016
04-03		Kyoto, Japan	2022	NIHS-DPP-60017
04-04		Mie, Japan	2022	NIHS-DPP-60018
04-05		Hyogo, Japan	2023	NIHS-DPP-60019
04-06		Nara, Japan	2021	NIHS-DPP-60020
05-01	Bupleurum root (柴胡)	Sichuan (四川), China	2010	NIHS-DPP-60021
06-01	Cornus fruit (山茱萸)	Shaanxi (陝西), China	Unknown	NIHS-DPP-60022
07-01	Peony root (芍薬)	Sichuan (四川), China	2018	NIHS-DPP-60023
08-01	Senna leaf (番瀉葉)	India	Unknown	NIHS-DPP-60024
09-01	Swertia herb(当薬)	Kochi, Japan	2021	NIHS-DPP-60025
10-01	Atractylodes lancea rhizome (蒼朮)	Shaanxi (陝西), China	2022	NIHS-DPP-60026
11-01	Rhubarb (大黄)	Sichuan (四川), China	2019	NIHS-DPP-60027
12-01	Alisma tuber (沢瀉)	Unknown	Unknown	NIHS-DPP-60028
13-01	Citrus unshiu peel (陳皮)	Kagawa, Japan	Unknown	NIHS-DPP-60029
14-01	Japanese angelica root (当帰)	Kyoto/Yamaguchi, Japan	2021, 2022	NIHS-DPP-60030
15-01	Peach kernel (桃仁)	Shanxi (山西), China	Unknown	NIHS-DPP-60031
16-01	Mentha herb (薄荷)	Zhejiang (浙江), China	Unknown	NIHS-DPP-60032
16-02		Anhui (安徽), China	2019	NIHS-DPP-60033
16-03		Anhui (安徽), China	2018	NIHS-DPP-60034
16-04		Anhui (安徽), China	2020	NIHS-DPP-60035
16-05		Zhejiang (浙江), China	2021	NIHS-DPP-60036
16-06		Shanghai (上海), China	2020	NIHS-DPP-60037
16-07		Anhui (安徽), China	Unknown	NIHS-DPP-60038
16-08		Shanghai (上海), China	2020	NIHS-DPP-60039
16-09[a]		Japan	2024	NIHS-DPP-60040
17-01	Pinellia tuber (半夏)	Gansu (甘肅), China	2021	NIHS-DPP-60041
18-01	Poria sclerotium (茯苓)	Unknown	Unknown	NIHS-DPP-60042
18-02		Yunnan (雲南), China	2019, 2021	NIHS-DPP-60043
18-03		Unknown	Unknown	NIHS-DPP-60044
18-04		China	2021	NIHS-DPP-60045
18-05		Yunnan (雲南), China	2022	NIHS-DPP-60046
18-06		Yunnan (雲南), China	2020	NIHS-DPP-60047
18-07		Nagano, Japan	2022	NIHS-DPP-60048
18-08		Liaoning (遼寧), China	2019	NIHS-DPP-60049
18-09		Jilin (吉林), China	2020	NIHS-DPP-60050

Table 1 (continued)

Sample ID	Sample name	Collection sites	Collection year	Specimen No.
18-10		Korea	2009	NIHS-DPP-60051
19-01	Ephedra herb (麻黄)	Neimenggu (内蒙古), China	2021	NIHS-DPP-60052
20-01	Coix seed (薏苡仁)	Thailand	2021	NIHS-DPP-60053
21-01	Oriental bezoar (牛黄)	Brazil	2020	NIHS-DPP-60054
22-01	Oyster shell (牡蛎)	Japan	2016	NIHS-DPP-60055
22-02		Hiroshima, Japan	2022	NIHS-DPP-60056
22-03		Seto Inland Sea, Japan	Unknown	NIHS-DPP-60057
22-04		Seto Inland Sea, Japan	2022	NIHS-DPP-60058
22-05		Seto Inland Sea, Japan	2021	NIHS-DPP-60059
22-06		Hiroshima, Japan	2021	NIHS-DPP-60060
22-07		Seto Inland Sea, Japan	2020	NIHS-DPP-60061
22-08		Unknown	Unknown	NIHS-DPP-60062
22-09		Liaoning (遼寧), China	2022	NIHS-DPP-60063
22-10[b]		China	2024	NIHS-DPP-60064
22-11[c]		China	2024	NIHS-DPP-60065
22-12[d]		China	2024	NIHS-DPP-60066
22-13[e]		China	2024	NIHS-DPP-60067
23-01	Longgu (竜骨)	Shaanxi (陜西), China	2023	NIHS-DPP-60068
23-02		Unknown	Unknown	NIHS-DPP-60069
23-03		Unknown	Unknown	NIHS-DPP-60070
23-04		Gansu (甘粛)/Ningxiahuizu (寧夏), China	2021	NIHS-DPP-60071
23-05		Gansu (甘粛), China	2023	NIHS-DPP-60072
23-06		Shaanxi (陜西), China	2021	NIHS-DPP-60073
23-07		Gansu (甘粛), China	2013	NIHS-DPP-60074
23-08		Shaanxi (陜西), China	2022	NIHS-DPP-60075
24-01	Aluminum silicate hydrate with silicon dioxide (滑石)	Fujian (福建), China	2015	NIHS-DPP-60076
25-01	Gypsum (石膏)	Shandong (山東), China	2021, 2022	NIHS-DPP-60077

[a]Cultivated in planters indoors and dried in the laboratory as a research sample

[b–e]Provided by the manufacturer as research samples

Table 2 EDXRF methods used in this study

	Quantitative analysis using calibration curves	Simultaneous multi-elemental analysis
Elements	$_{24}$Cr, $_{29}$Cu, $_{33}$As, $_{48}$Cd, $_{80}$Hg, $_{82}$Pb	$_{13}$Al–$_{92}$U
Detector	Silicon drift detector	Silicon drift detector
X-Ray tube	Rh target	Rh target
Tube current	Auto	Auto
Tube voltage	50 kV	15 kV (S–K, Na–Sc) 50 kV (Al–U, Cr–Fe, Zn–As, Pb, Rh–Cd)
Primary filter	#2 (Cr), #4 (Cu, As, Hg, Pb), #1 (Cd)	None (Al–U, Na–Sc), #2 (S–K), #3 (Cr–Fe), #4 (Zn–As, Pb), #1 (Rh–Cd)
Collimator	10 mm diameter	10 mm diameter
Atmosphere	Air	Air
Integration time	1800s × 3 Channel	300 s × 6 Channel
Dead time	Max. 30%	Max. 30%
Other	–	A balance setting for fundamental parameter method: CH_2O

polypropylene film on the bottom and subsequently pressed with a stick.

ICP-MS analytical samples: Sample powder (0.2 g) was mixed with 5 ml of nitric acid in a fluoropolymer vessel and microwaved at 180 °C for 10 min. After cooling, 1 ml of Au solution (5 mg/l) containing 0.1 ml of nitric acid, along with water, was added to reach a total volume of 50 ml.

Quantitative analyses via the calibration curve method using EDXRF and ICP-MS

EDXRF analysis: Calibration standards were prepared using an appropriate amount of water in the following concentration ranges: Cr, 0–50 ppm; Cu, 0–200 ppm; As, 0–15 ppm; Cd, 0–5 ppm; Hg, 0–30 ppm; and Pb, 0–50 ppm. These standards were analyzed, and calibration curves were created. The calibration curves were applied with shape and material correction using the scattering X-rays, and the As calibration curve was corrected using the dj method for overlapping Pb. Each sample was analyzed three times using the calibration curves (Sample ID: 21-01 was analyzed twice owing to its small amount), and the average value is shown as the quantitative value. Samples were prepared separately for each analysis. Furthermore, the limit of quantitation (LOQ; 10σ) was calculated based on the limit of detection (LOD; 3σ) indicated on the PCEDX-Pro software. Quantitation values below the LOD were judged as not detected (N.D.) and those below the LOQ as Trace (Tr.).

ICP-MS analysis: Calibration standards were prepared using an Au solution (0.1 mg/l), including nitric acid (0.1 ml/ml) in the following concentration ranges: Cr, 0–30 ppb; Cu, 0–30 ppb; As, 0–12 ppb; Cd, 0–1.8 ppb; Hg, 0–1.2 ppb; and Pb, 0–30 ppb. These standards were analyzed, and calibration curves were created. Each sample was analyzed twice based on the calibration curves, and the average value was used as the quantitative value. Samples were prepared separately for each analysis. The LOD (3σ) and LOQ (10σ) were calculated from the standard deviation (σ) obtained from six repeated measurements of the same standard solution. Quantitation values below the LOD were judged as N.D., whereas those below the LOQ were judged as Tr.

Simultaneous multi-elemental analysis using EDXRF

Simultaneous multi-elemental analysis was performed on Coptis rhizome (Sample ID: 02-01 to 12), Brown rice (Sample ID: 04-01 to 06), Mentha herb (Sample ID: 16-01 to 09), Poria sclerotium (Sample ID: 18-01 to 10), Oyster shell (Sample ID: 22-01 to 13), and Longgu (Sample ID: 23-01 to 08) using EDXRF (Table 1). The measurement conditions are listed in Table 2. In the X-ray fluorescence spectra, the energy position of the peak is recorded to identify the elements. Elements above the LOD (3σ) are judged as detected based on the fundamental parameter method. Each elemental content is considered by superimposing the X-ray fluorescence spectra of the same herbal medicine and comparing the intensity of the same elemental peak.

Results

Quantitative results of EDXRF and ICP-MS calibration curves and their comparison

EDXRF and ICP-MS calibration curves for Cr, Cu, As, Cd, Hg, and Pb showed good linearity, with a correlation coefficient (R) ≥ 0.99. The EDXRF and ICP-MS calibration curves are presented in Figs. 1 and 2, respectively. Quantitative EDXRF and ICP-MS analyses are performed for 25 types of herbal medicines using these calibration curves. The LOQ of EDXRF and ICP-MS is listed in Table 3.

The EDXRF LOQ tended to be higher for animals and minerals (Sample ID: 21-01 to 25-01) than for plants (Sample ID: 01-01 to 20-01). EDXRF and ICP-MS quantitative values based on these LOQ are listed in Table 4. The detailed quantitative results for each element are shown below.

Cr

EDXRF quantitative values of all 25 herbal medicines were below the LOQ (1.14–2.84 ppm). However, ICP-MS enabled the detection of 0.02–1.46 ppm in all 25 herbal medicines. Among these, only two herbal medicines exceeded 1 ppm: Peach kernel (1.46 ppm) and Oriental bezoar (1.05 ppm).

Cu

EDXRF detected 1.04–139.13 ppm in 22 herbal medicines: except Longgu, Aluminum silicate hydrate with silicon dioxide, and Gypsum; these were below the LOQ (0.47–0.63 ppm). In contrast, ICP-MS detected 0.57–129.78 ppm in 22 herbal medicines: except Oyster shell, Aluminum silicate hydrate with silicon dioxide, and Gypsum, which were below the LOQ (0.54 ppm). The ratio of the EDXRF and ICP-MS quantitative values was calculated for comparison. Both values were in good agreement.

As

EDXRF detected 0.33–1.14 ppm in four herbal medicines: Chrysanthemum flower, Atractylodes lancea rhizome, Mentha herb, and Oriental bezoar. Each ratio of the EDXRF quantitative value to ICP-MS was 133–198%; all ICP-MS quantitative values were < 1 ppm, likely resulting in an increase in the ratios. The ICP-MS quantitative values were markedly low (0.009–0.250 ppm) for the 21 herbal

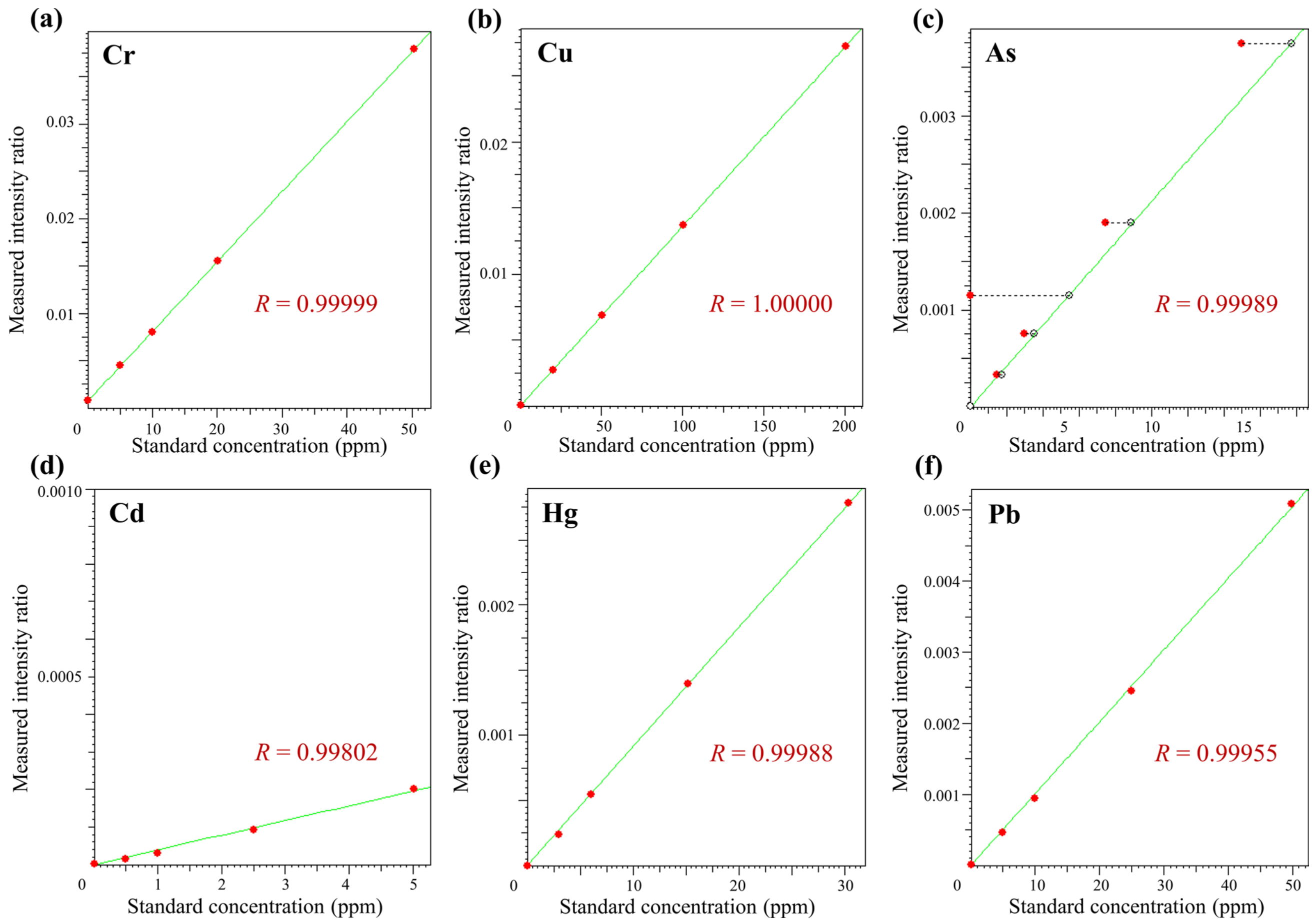


Fig. 1 EDXRF calibration curves of Cr (**a**), Cu (**b**), As (**c**), Cd (**d**), Hg (**e**), and Pb (**f**)

medicines that were below the EDXRF LOQ. This indicates that the results of both methods can be explained without contradictions.

Cd

The EDXRF quantitative values of all 25 herbal medicines were below the LOQ (1.08–1.89 ppm). However, ICP-MS enabled the detection of 0.003–1.418 ppm in 24 herbal medicines, with the exception of Gypsum, which is below the LOQ (0.0007 ppm). Among these, only the Chrysanthemum flower (1.418 ppm) exceeded 1 ppm.

Hg

EDXRF quantitative values of all 25 herbal medicines were below the LOQ (0.13–0.54 ppm). Although ICP-MS enabled the detection of 0.039 ppm in Oriental bezoar, the other 24 herbal medicines were below the LOQ (0.012 ppm). An overlapping peak that appeared to be Ge near the Hg peak was detected via the Aluminum silicate hydrate with silicon dioxide, and analytical processing was performed to correct the overlap.

Pb

EDXRF detected 0.60–10.54 ppm in six herbal medicines: Coptis rhizome, Chrysanthemum flower, Bupleurum root, Mentha herb, Oriental bezoar, and Longgu. These quantitative values are in good agreement with ICP-MS results. In contrast, Aluminum silicate hydrate with silicon dioxide showed a substantial difference between the EDXRF and ICP-MS quantitative values, with a ratio of 183%. The other 18 herbal medicines that were below the LOQ (0.27–0.88 ppm) in EDXRF analysis also exhibited considerably low quantitative values (< 0.30 ppm) during ICP-MS.

Comparison of EDXRF quantitative results of herbal medicines in different collection sites or lots

To investigate the variations in heavy metal content among the same herbal medicines, multiple samples with different collection sites or lots were subjected to quantitative

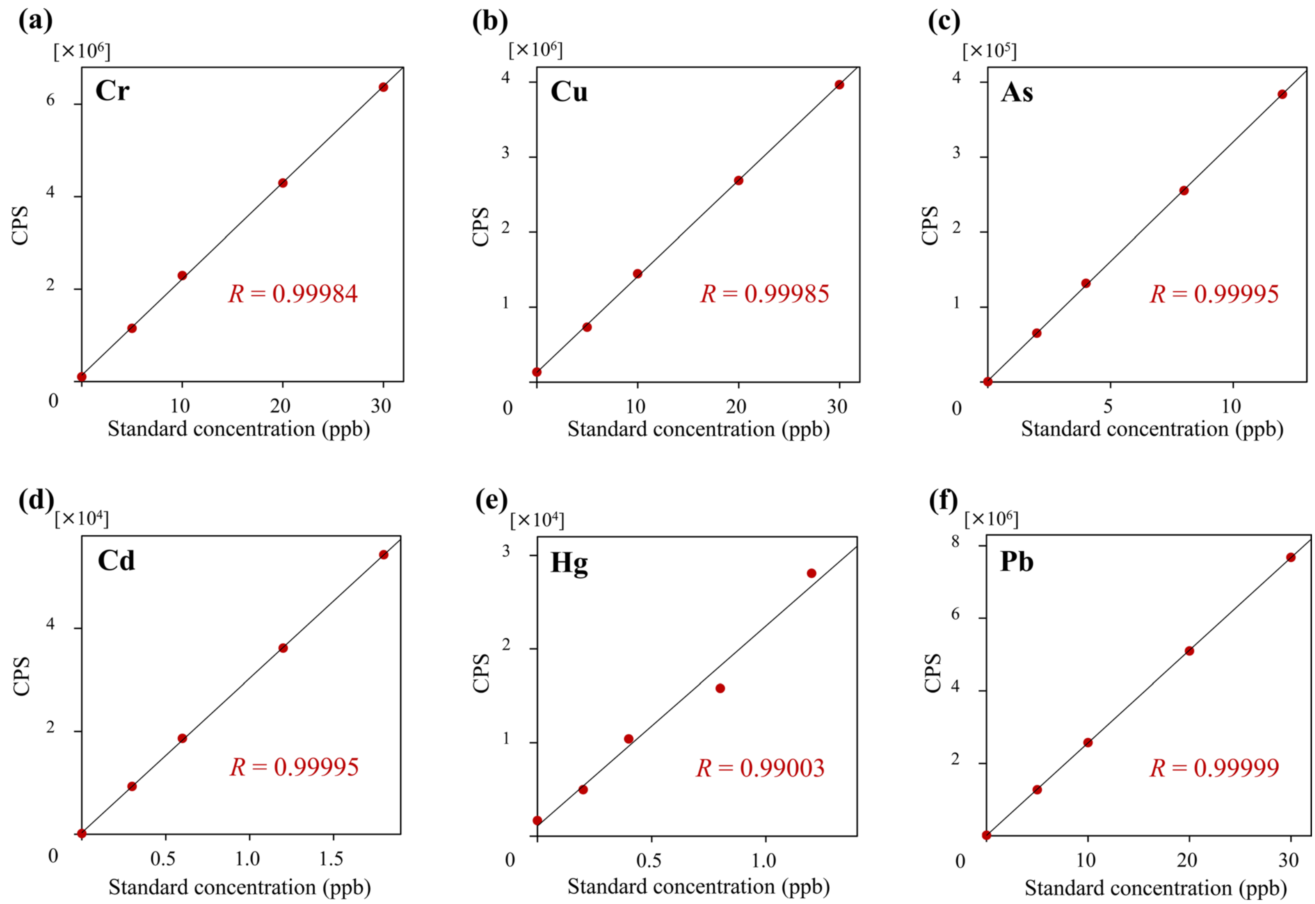


Fig. 2 ICP-MS calibration curves of Cr (**a**), Cu (**b**), As (**c**), Cd (**d**), Hg (**e**), and Pb (**f**)

analysis of six elements (Cr, Cu, As, Cd, Hg, and Pb) using the EDXRF calibration curve method. For this experiment, six herbal medicines using different parts and materials were selected: Coptis rhizome, Brown rice, Mentha herb, Poria sclerotium, Oyster shell, and Longgu. Comprehensive quantitative results for each herbal medicine are shown below:

Coptis rhizome

Table 5 presents the results for this herbal medicine. The Cr, As, Cd, and Hg contents were below the LOQ in all 12 samples. Cu contents were 9.76–23.87 ppm in all samples, and samples collected in Japan showed variation. Pb was detected in 11 samples, with the exception of one sample (Sample ID: 02-10). Samples collected in China showed similar contents of 0.65–1.63 ppm, whereas those in Japan showed a large variation (0.35–27.36 ppm).

Brown rice

Table 6 presents the results for this herbal medicine. The Cr, Cd, Hg, and Pb contents were below the LOQ in all six samples. Cu contents were 3.07–3.89 ppm in all samples, with minimal variety observed. As contents were approximately 0.4 ppm in two samples (Sample ID: 04-04, 04-05).

Mentha herb

Table 7 presents the results for this herbal medicine. The Cd and Hg contents were below the LOQ for all nine samples. The samples collected in China contained Cu, Pb, Cr, and As contents of 10.79–13.77 ppm, 0.39–3.13 ppm, 1.55–1.88 ppm, and 0.33–0.59 ppm, respectively. Cu was detected at 2.31 ppm, whereas the Cr, As, and Pb contents were all below the LOQ in one sample collected in Japan (Sample ID: 16-09).

Poria sclerotium

Table 8 presents the results for this herbal medicine. The Cr, As, Cd, and Hg contents were below the LOQ in all 10 samples. Cu contents were 1.47–3.04 ppm in all samples, and Pb was detected at 0.35 ppm in one sample (Sample ID: 18-09).

Table 3 The limits of quantitation (LOQ) for EDXRF and ICP-MS [ppm]

	Sample ID	Sample name	Cr	Cu	As	Cd	Hg	Pb
EDXRF	01-01	Phellodendron bark (黄柏)	1.58	0.28	0.25	1.21	0.17	0.32
	02-01	Coptis rhizome (黄連)	1.15	0.44	0.24	1.08	0.13	0.29
	03-01	Chrysanthemum flower (菊花)	1.34	0.48	0.27	1.27	0.17	0.32
	04-01	Brown rice (粳米)	1.18	0.26	0.22	1.12	0.13	0.28
	05-01	Bupleurum root (柴胡)	1.25	0.38	0.23	1.23	0.13	0.28
	06-01	Cornus fruit (山茱萸)	1.31	0.28	0.24	1.28	0.16	0.29
	07-01	Peony root (芍薬)	1.24	0.31	0.22	1.09	0.15	0.28
	08-01	Senna leaf (番瀉葉)	1.60	0.36	0.26	1.23	0.18	0.33
	09-01	Swertia herb (当薬)	1.45	0.34	0.22	1.40	0.14	0.28
	10-01	Atractylodes lancea rhizome (蒼朮)	1.33	0.40	0.24	1.24	0.17	0.29
	11-01	Rhubarb (大黄)	1.34	0.31	0.24	1.29	0.17	0.29
	12-01	Alisma tuber (沢瀉)	1.21	0.47	0.22	1.14	0.14	0.27
	13-01	Citrus unshiu peel (陳皮)	1.20	0.24	0.21	1.20	0.15	0.27
	14-01	Japanese angelica root (当帰)	1.33	0.34	0.25	1.20	0.17	0.31
	15-01	Peach kernel (桃仁)	1.15	0.37	0.21	1.10	0.13	0.28
	16-01	Mentha herb (薄荷)	1.40	0.47	0.30	1.24	0.18	0.34
	17-01	Pinellia tuber (半夏)	1.21	0.27	0.21	1.22	0.15	0.27
	18-01	Poria sclerotium (茯苓)	1.14	0.23	0.21	1.18	0.13	0.28
	19-01	Ephedra herb (麻黄)	1.32	0.28	0.25	1.32	0.15	0.32
	20-01	Coix seed (薏苡仁)	1.20	0.26	0.21	1.09	0.14	0.27
	21-01	Oriental bezoar (牛黄)	1.28	1.36	0.41	1.31	0.18	0.43
	22-01	Oyster shell (牡蛎)	2.84	0.63	0.59	1.65	0.47	0.67
	23-01	Longgu (竜骨)	2.41	0.63	0.59	1.89	0.54	0.63
	24-01	Aluminum silicate hydrate with silicon dioxide (滑石)	2.52	0.47	0.87	1.55	0.46	0.88
	25-01	Gypsum (石膏)	2.01	0.53	0.48	1.59	0.40	0.54
ICP-MS			0.01	0.54	0.001	0.001	0.01	0.003

Oyster shell

Table 9 presents the results for this herbal medicine. The As, Cd, Hg, and Pb contents were below the LOQ in all 13 samples. Cr was detected at 3.49 ppm in one sample (Sample ID: 22-12), and Cu contents were 0.85–1.69 ppm in all samples.

Longgu

Table 10 presents the results for this herbal medicine. The Cd and Hg contents were below the EDXRF LOQ for all eight samples. EDXRF detected 2.65–12.29 ppm of Cr in seven samples and 1.97–3.65 ppm of Cu in six samples, along with 2.16–9.73 ppm of As and 0.92–10.00 ppm of Pb in all samples. Some samples showed an overlapping peak that seemed to be Ba near Cr or one that seemed to be Sr near Pb, and analytical processing was performed to correct the overlaps.

Considering that overlapping peaks may affect the EDXRF quantitative values, ICP-MS quantitative analysis was performed on all eight samples and compared with the EDXRF results. The ratios of the EDXRF quantitative values to the ICP-MS are listed in Table 10. The quantitative EDXRF values tended to be slightly higher than those of ICP-MS.

Simultaneous multi-elemental analysis using EDXRF

Simultaneous multi-elemental analysis using EDXRF was performed on the six herbal medicines analyzed in the previous experiment to determine whether other elemental contents, except for the six elements (Cr, Cu, As, Cd, Hg, and Pb), varied. The detected elements are listed in Table 11. Twenty-two elements, including four elements (Cr, Cu, As, and Pb) targeted in the calibration curve method, were detected. Among these, five elements (Ca, Mn, Fe, Cu, and Zn) were commonly detected in the six herbal medicines. However, the detection trends for other elements differed. The comprehensive results for each herbal medicine are shown below.

Table 4 EDXRF and ICP-MS quantitative values of the 25 herbal medicines [ppm]

Sample ID	Sample name	Cr		Cu			As			Cd		Hg		Pb		
		EDX	ICP	EDX	ICP	Ratio (%)[a]	EDX	ICP	Ratio (%)	EDX	ICP	EDX	ICP	EDX	ICP	Ratio (%)
01-01	Phellodendron bark (黄柏)	Tr.[b]	0.41	2.19	2.69	81	N.D.[c]	0.009	–[d]	N.D.	0.014	N.D.	N.D.	N.D.	0.22	–
02-01	Coptis rhizome (黄連)	N.D.	0.42	16.67	16.56	101	Tr.	0.044	–	Tr.	0.437	N.D.	N.D.	1.25	1.63	77
03-01	Chrysanthemum flower (菊花)	Tr.	0.86	14.42	15.14	95	0.35	0.194	180	Tr.	1.418	N.D.	N.D.	0.97	1.25	78
04-01	Brown rice (粳米)	N.D.	0.23	3.07	4.48	69	Tr.	0.250	–	N.D.	0.041	N.D.	N.D.	N.D.	0.02	–
05-01	Bupleurum root (柴胡)	N.D.	0.67	10.52	11.51	91	Tr.	0.095	–	Tr.	0.756	N.D.	N.D.	0.60	0.85	71
06-01	Cornus fruit (山茱萸)	Tr.	0.51	1.74	1.89	92	N.D.	0.045	–	N.D.	0.009	N.D.	N.D.	Tr.	0.14	–
07-01	Peony root (芍薬)	N.D.	0.30	4.59	5.26	87	Tr.	0.032	–	Tr.	0.538	N.D.	N.D.	N.D.	0.08	–
08-01	Senna leaf (番瀉葉)	Tr.	0.74	5.02	5.91	85	Tr.	0.101	–	N.D.	0.020	N.D.	N.D.	N.D.	0.13	–
09-01	Swertia herb (当薬)	Tr.	0.69	7.19	7.53	95	Tr.	0.094	–	N.D.	0.268	N.D.	N.D.	Tr.	0.30	–
10-01	Atractylodes lancea rhizome (蒼朮)	N.D.	0.57	8.48	8.34	102	0.33	0.242	136	N.D.	0.230	N.D.	N.D.	Tr.	0.39	–
11-01	Rhubarb (大黄)	N.D.	0.46	3.22	3.47	93	Tr.	0.109	–	N.D.	0.034	N.D.	N.D.	Tr.	0.12	–
12-01	Alisma tuber (沢瀉)	N.D.	0.47	16.43	17.14	96	Tr.	0.184	–	N.D.	0.482	N.D.	N.D.	Tr.	0.16	–
13-01	Citrus unshiu peel (陳皮)	N.D.	0.37	1.39	1.63	85	N.D.	0.015	–	N.D.	0.003	N.D.	N.D.	N.D.	0.05	–
14-01	Japanese Angelica root (当帰)	Tr.	0.76	4.76	4.77	100	Tr.	0.089	–	N.D.	0.105	N.D.	N.D.	N.D.	0.13	–
15-01	Peach kernel (桃仁)	N.D.	1.46	10.29	10.43	99	N.D.	0.013	–	N.D.	0.009	N.D.	N.D.	N.D.	0.02	–
16-01	Mentha herb (薄荷)	Tr.	0.92	12.66	12.87	98	0.51	0.257	198	N.D.	0.043	N.D.	Tr.	2.38	2.76	86
17-01	Pinellia tuber (半夏)	N.D.	0.18	3.03	2.95	103	N.D.	0.031	–	N.D.	0.031	N.D.	N.D.	N.D.	0.02	–
18-01	Poria sclerotium (茯苓)	N.D.	0.20	2.00	2.51	80	Tr.	0.027	–	N.D.	0.016	N.D.	N.D.	N.D.	0.04	–
19-01	Ephedra herb (麻黄)	N.D.	0.65	2.03	1.96	104	Tr.	0.133	–	N.D.	0.038	N.D.	N.D.	N.D.	0.15	–
20-01	Coix seed (薏苡仁)	N.D.	0.38	2.52	2.82	89	N.D.	0.019	–	N.D.	0.004	N.D.	N.D.	N.D.	0.01	–
21-01	Oriental bezoar (牛黄)	N.D.	1.05	139.13	129.78	107	1.14	0.856	133	N.D.	0.035	N.D.	0.039	10.54	11.41	92
22-01	Oyster shell (牡蛎)	Tr.	0.12	1.04	Tr.	–	N.D.	0.059	–	N.D.	0.007	N.D.	N.D.	Tr.	0.18	–
23-01	Longgu (竜骨)	Tr.	0.15	Tr.	0.57	–	N.D.	0.057	–	N.D.	0.004	N.D.	N.D.	0.92	0.82	112
24-01	Aluminum silicate hydrate with silicon dioxide (滑石)	N.D.	0.02	Tr.	N.D.	–	N.D.	0.142	–	N.D.	0.048	N.D.	N.D.	30.60	16.68	183
25-01	Gypsum (石膏)	Tr.	0.02	Tr.	N.D.	–	N.D.	0.010	–	N.D.	Tr.	N.D.	N.D.	N.D.	Tr.	–

[a]Ratio of EDXRF quantitative values to ICP-MS values (%)

[b]Below the LOQ (10σ)

[c]Below the LOD (3σ)

[d]Not available

Table 5 EDXRF quantitative values of Coptis rhizome (黄連) [ppm]

Sample ID	Collection sites (Country)	Cr	Cu	As	Cd	Hg	Pb
02-01	China	N.D.[a]	16.67	Tr.[b]	Tr.	N.D.	1.25
02-02	China	N.D.	17.07	Tr.	N.D.	N.D.	1.14
02-03	China	N.D.	17.28	Tr.	Tr.	N.D.	0.65
02-04	China	N.D.	17.09	Tr.	N.D.	N.D.	1.09
02-05	China	N.D.	19.22	Tr.	Tr.	N.D.	1.27
02-06	China	N.D.	16.94	Tr.	Tr.	N.D.	1.29
02-07	China	N.D.	17.22	Tr.	N.D.	N.D.	1.20
02-08	China	Tr.	20.95	0.30	Tr.	N.D.	1.63
02-09	Japan	N.D.	23.87	Tr.	N.D.	N.D.	0.35
02-10	Japan	N.D.	9.76	Tr.	Tr.	N.D.	Tr.
02-11	Japan	N.D.	12.76	1.04	N.D.	N.D.	24.59
02-12	Japan	N.D.	12.58	1.27	N.D.	N.D.	27.36
LOQ		1.20	0.46	0.28	1.13	0.15	0.33

[a]Below the LOD (3σ)

[b]Below the LOQ (10σ)

Table 6 EDXRF quantitative values of Brown rice (粳米) [ppm]

Sample ID	Collection sites (Country)	Cr	Cu	As	Cd	Hg	Pb
04-01	Japan	N.D.[a]	3.07	Tr.[b]	N.D.	N.D.	N.D.
04-02	Japan	N.D.	3.10	Tr.	N.D.	N.D.	N.D.
04-03	Japan	N.D.	3.74	Tr.	N.D.	N.D.	N.D.
04-04	Japan	N.D.	3.46	0.36	N.D.	N.D.	N.D.
04-05	Japan	N.D.	3.89	0.39	N.D.	N.D.	N.D.
04-06	Japan	N.D.	3.13	Tr.	N.D.	N.D.	Tr.
LOQ		1.19	0.27	0.21	1.13	0.14	0.27

[a]Below the LOD (3σ)

[b]Below the LOQ (10σ)

Table 7 EDXRF quantitative values of Mentha herb (薄荷) [ppm]

Sample ID	Collection sites (Country)	Cr	Cu	As	Cd	Hg	Pb
16-01	China	Tr.[a]	12.66	0.51	N.D.[b]	N.D.	2.38
16-02	China	Tr.	11.48	0.33	N.D.	N.D.	0.39
16-03	China	Tr.	12.21	0.41	N.D.	N.D.	3.13
16-04	China	Tr.	13.77	0.47	N.D.	N.D.	1.99
16-05	China	Tr.	13.23	0.49	N.D.	N.D.	2.37
16-06	China	1.88	11.68	0.39	N.D.	N.D.	1.01
16-07	China	Tr.	10.79	0.59	N.D.	N.D.	2.93
16-08	China	1.55	11.55	0.36	N.D.	N.D.	0.76
16-09[c]	Japan	Tr.	2.31	Tr.	N.D.	N.D.	N.D.
LOQ		1.41	0.45	0.29	1.33	0.18	0.33

[a]Below the LOQ (10σ)

[b]Below the LOD (3σ)

[c]Cultivated in planters indoors and dried in the laboratory as a research sample

Table 8 EDXRF quantitative values of Poria sclerotium (茯苓) [ppm]

Sample ID	Collection sites (Country)	Color	Cr	Cu	As	Cd	Hg	Pb
18-01	China	White	N.D.[a]	2.00	Tr.[b]	N.D.	N.D.	N.D.
18-02	China	White	N.D.	2.76	N.D.	N.D.	N.D.	N.D.
18-03	China	White	N.D.	2.65	N.D.	N.D.	N.D.	N.D.
18-04	China	White	N.D.	3.03	N.D.	N.D.	N.D.	N.D.
18-05	China	White	N.D.	3.04	N.D.	N.D.	N.D.	N.D.
18-06	China	White	N.D.	2.56	N.D.	N.D.	N.D.	Tr.
18-07	China	Red	N.D.	1.73	N.D.	N.D.	N.D.	N.D.
18-08	China	Red	N.D.	1.94	N.D.	N.D.	N.D.	Tr.
18-09	China	Red	N.D.	1.63	Tr.	N.D.	N.D.	0.35
18-10	China	Red	N.D.	1.47	N.D.	N.D.	N.D.	Tr.
LOQ			1.15	0.24	0.21	1.19	0.13	0.27

[a]Below the LOD (3σ)

[b]Below the LOQ (10σ)

Table 9 EDXRF quantitative values of Oyster shell (牡蛎) [ppm]

Sample ID	Collection sites (country)	Cr	Cu	As	Cd	Hg	Pb
22-01	Japan	Tr.[a]	1.04	N.D.[b]	N.D.	N.D.	Tr.
22-02	Japan	Tr.	1.33	N.D.	N.D.	N.D.	Tr.
22-03	Japan	Tr.	1.26	N.D.	N.D.	N.D.	Tr.
22-04	Japan	Tr.	1.22	N.D.	N.D.	N.D.	Tr.
22-05	Japan	Tr.	1.24	N.D.	N.D.	N.D.	Tr.
22-06	Japan	Tr.	0.85	N.D.	Tr.	N.D.	Tr.
22-07	Japan	Tr.	1.14	N.D.	N.D.	N.D.	Tr.
22-08	Unknown	Tr.	1.08	N.D.	Tr.	N.D.	Tr.
22-09	China	Tr.	0.89	N.D.	N.D.	N.D.	Tr.
22-10[c]	China	Tr.	1.69	Tr.	Tr.	N.D.	Tr.
22-11[d]	China	Tr.	1.43	Tr.	N.D.	N.D.	Tr.
22-12[e]	China	3.49	1.23	N.D.	N.D.	N.D.	Tr.
22-13[f]	China	Tr.	1.35	N.D.	N.D.	N.D.	Tr.
LOQ		2.94	0.66	0.61	1.72	0.51	0.69

[a]Below the LOQ (10σ)

[b]Below the LOD (3σ)

[c–f]Provided by the manufacturer as research samples

Coptis rhizome

Representative X-ray fluorescence spectra of Coptis rhizome are shown in Fig. S1. In total, 15 elements were detected: Si, P, S, Cl, K, Ca, Mn, Fe, Cu, Zn, Br, Rb, Sr, Ba, and Pb. The Japanese samples tended to contain more Zn and less Mn and Fe than the Chinese samples.

Brown rice

The representative X-ray fluorescence spectra of Brown rice are shown in Fig. S2. Eleven elements were detected: P, S, Cl, K, Ca, Mn, Fe, Cu, Zn, Br, and Rb. All the samples showed similar spectra.

Mentha herb

Representative X-ray fluorescence spectra of Mentha herb are shown in Fig. S3. In total, 16 elements were detected: Si, P, S, Cl, K, Ca, Ti, Mn, Fe, Cu, Zn, Br, Rb, Sr, Ba, and Pb. Ti was a characteristic element of Mentha herb. Chinese samples tended to contain more of seven elements (S, Ca, Fe, Cu, Zn, Sr, and Pb) than the Japanese sample. Of these, Ca, Fe, Cu, and Sr differed by more than two-fold. In contrast,

Table 10 EDXRF and ICP-MS quantitative values of Longgu (竜骨) [ppm]

Sample ID	Cr			Cu			As			Cd		Hg		Pb		
	EDX	ICP	Ratio (%)[a]	EDX	ICP	Ratio (%)	EDX	ICP	Ratio (%)	EDX	ICP	EDX	ICP	EDX	ICP	Ratio (%)
23-01	Tr.[b]	0.15	–[c]	Tr.	0.59	–	N.D.[d]	0.06	–	N.D.	0.004	N.D.	N.D.	0.92	0.85	112
23-02	12.09	11.41	106	3.65	3.69	99	7.10	6.02	118	Tr.	0.161	Tr.	0.036	10.00	7.92	126
23-03	2.65	1.65	161	Tr.	0.69	–	2.16	1.84	117	Tr.	0.022	N.D.	N.D.	1.78	1.42	125
23-04	4.50	6.92	65	2.23	2.06	108	8.35	7.16	117	N.D.	0.083	N.D.	N.D.	6.38	5.00	128
23-05	7.11	5.12	139	2.63	2.11	125	8.78	7.17	122	Tr.	0.086	N.D.	N.D.	6.50	4.55	143
23-06	6.65	4.67	142	1.97	1.44	137	5.73	4.29	134	Tr.	0.067	N.D.	N.D.	4.76	3.01	158
23-07	4.83	6.93	70	2.76	2.02	137	9.73	7.07	138	Tr.	0.080	N.D.	N.D.	7.68	5.10	151
23-08	12.29	10.24	120	2.40	1.85	130	5.80	3.90	149	Tr.	0.189	Tr.	N.D.	6.42	3.91	164
LOQ	2.41–3.05	0.01	–	0.63–0.72	0.54	–	0.59–0.90	0.001	–	1.75–0.93	0.001	0.54–0.61	0.01	0.63–0.80	0.003	–

[a]Ratio of EDXRF quantitative values to ICP-MS values (%)

[b]Below the LOQ (10σ)

[c]Not available

[d]Below the LOD (3σ)

Table 11 Elements detected in simultaneous multi-elemental analyses using EDXRF

Sample name	14Si	15P	16S	17Cl	19K	20Ca	22Ti	24Cr	25Mn	26Fe	29Cu	30Zn	33As	35Br	37Rb	38Sr	39Y	53I	56Ba	62Sm	82Pb	92U
Coptis rhizome (黄連)	+[a]	+	+	+	+	+			+	+	+	+		+	+	+			+		+	
Brown rice (粳米)		+	+	+	+	+			+	+	+	+		+	+							
Mentha herb (薄荷)	+	+	+	+	+	+	+		+	+	+	+		+	+	+			+		+	
Poria sclerotium (茯苓)		+	+		+	+			+	+	+	+		+	+	+						
Oyster shell (牡蛎)			+	+		+			+	+	+	+		+		+				+		
Longgu (竜骨)	+	+		+	+	+		+	+	+	+	+	+			+	+	+	+		+	+

[a]Detected

the Japanese sample contained slightly more Cl and K than Chinese samples.

Poria sclerotium

The representative X-ray fluorescence spectra of Poria sclerotium are shown in Fig. S4. Eleven elements were detected: P, S, K, Ca, Mn, Fe, Cu, Zn, Br, Rb, and Sr. Red-looking Poria sclerotium (known as "Red Poria") tended to contain more Ca and less of five elements (S, K, Mn, Fe, and Zn) than its white-looking counterpart (known as "White Poria"). The colors corresponding to these appearances are listed in Table 8.

Oyster shell

Representative X-ray fluorescence spectra of the Oyster shell are shown in Fig. S5. Ten elements were detected: S, Cl, Ca, Mn, Fe, Cu, Zn, Br, Sr, and Sm. Sm was a characteristic element in oyster shell. All the samples showed similar spectra.

Longgu

The representative X-ray fluorescence spectra of Longgu are shown in Fig. S6. In total, 17 elements were detected: Si, P, Cl, K, Ca, Cr, Mn, Fe, Cu, Zn, As, Sr, Y, I, Ba, Pb, and U. The contents of P and Ca, which are considered the main components, showed no significant differences. Contrastingly, Mn, Fe, Sr, and Ba contents exhibited considerable variation. I, Y, and U, which appear to be distinctive elements of Longgu, were not detected in one sample (Sample ID: 23-01).

Discussion

The EDXRF analysis of herbal medicines has been conducted [12, 13]. However, only a few EDXRF studies have investigated differences in the LOQ or coexisting elements for each herbal medicine. In addition, the suitability of EDXRF analysis for herbal medicines remains unclear. Therefore, we evaluated the ability of this method to compare the quantitative values of six elements (Cr, Cu, As, Cd, Hg, and Pb) in 25 herbal medicines using ICP-MS, which enabled highly sensitive and accurate quantitative analysis. We obtained quantitative EDXRF values in the ratio of 65–198% for ICP-MS (Tables 4 and 10). This suggests that EDXRF is effective for the quantitative analysis of heavy metals in herbal medicines, considering that the targeted elements were traces.

The Mentha herb showed the highest Pb content among the 20 plant samples (Sample ID: 01-01 to 20-01) in this study (Table 4). The Mentha herb collected in China (Sample ID: 16-01 to 16-08, cultivated outdoors) showed an average Pb content of 1.87 ppm (0.39–3.13 ppm), whereas that collected in Japan (Sample ID: 16-09, cultivated indoors) showed Pb content below the LOQ (Table 7). Dinu et al. [14] added metals during the cultivation of *Mentha* species to quantify the transfer of toxic metals in soil; the leaves accumulated lower amounts of metals than the roots. Contrastingly, the leaves of wild *Mentha* species grown under a considerably polluted environment [15] contained more toxic heavy metals than the stems, even after the removal of heavy metals attached to the surface. These studies indicate the importance of controlling the plant growth environment.

Coptis rhizome and Longgu have specified permissible heavy metal limits in JP. Coptis rhizome can effectively uptake heavy metals from the soil. Moreover, the Pb content of the fibrous roots and rhizomes of Coptis rhizome positively correlated with that in the growth soil [16]. The permissible Limit of heavy metals in JP Coptis rhizome is 20 ppm. In case where visual judgment is difficult owing to turbidity, it is necessary to determine the amount of Pb as < 5 ppm using atomic absorption spectrophotometry [1, 17]. Of the 12 Coptis rhizome samples in this study, only two collected in Japan (Sample ID: 02-11 and 02-12) contained Pb > 5 ppm (Table 5). Test solutions of the two samples were prepared following the procedure for JP Coptis rhizome labeled for use in extracts, infusions, and decoctions. They were subsequently analyzed using EDXRF; both Pb contents were below the LOD (0.10 ppm).

Longgu is a fossil defined as the ossified bone of a large mammal. Fossils are buried underground for long periods and are assumed to contain various trace elements derived from the Earth's crust and seawater. The permissible Limits of heavy metals and As in JP Longgu are 20 ppm and 10 ppm, respectively. EDXRF quantitative values of Longgu samples (Sample ID: 23-01 to 23-08; Table 10) were high; the Pb and As contents averaged 5.56 ppm (0.92–10.00 ppm) and 6.81 ppm (2.16–9.73 ppm; excluding data for N.D.), respectively. Test solutions of samples with the highest Pb (Sample ID: 23-02) and As (Sample ID: 23-07) contents were prepared following the procedure for JP Longgu labeled for use in extracts, infusions, and decoctions; they were then analyzed using EDXRF. The Pb (Sample ID: 23-02) and As (Sample ID: 23-07) contents were both below 0.10 ppm, and ultimately below the permissible Limit of 0.5 ppm. A lower heavy metal content May indicate a better qualified sample, such as Sample ID: 23-01. However, this sample did not contain elements that are unique to Longgu (I, Y, and U), which may indicate its questionable origin. Thus, EDXRF analysis can be used in combination with simultaneous multi-elemental analysis for the quality control and evaluation of herbal medicines.

The permissible Limits of the Heavy Metals Limit Test are 10–40 ppm for JP herbal medicines [1]. This test measures highly toxic elements (Cd, Sn, Hg, and Pb). However, it also reacts with other elements (Fe, Cu, Sb, and Bi) [17], which is a major disadvantage. In contrast, the EDXRF quantitative values were rarely affected by other elements. Of the elements that react in the JP Heavy Metals Limit Test, three (Cd, Hg, and Pb) are highly toxic and May be present in herbal medicines. The Maximum EDXRF LOQ in this study totaled 3.31 ppm for Cd (1.89 ppm), Hg (0.54 ppm), and Pb (0.88 ppm). This value is below the lowest permissible Limit of 10 ppm for JP herbal medicines. Even if all quantitative values of the three elements determined using EDXRF are below the LOQ and do not show specific values, the total contents are estimated to be below the permissible limits for JP herbal medicines. Similarly, the maximum EDXRF LOQ of As (0.87 ppm) was below the lowest permissible Limit of 2 ppm (2–10 ppm) for JP herbal medicines in the JP Arsenic Limit Test. The LOQ in this study suggests that EDXRF enables the determination of heavy metals more accurately than the JP tests. However, this study had two limitations: one is the small size of animal samples. Therefore, EDXRF analysis on more animal samples is required to confirm its applicability to herbal medicines. The other is that most of the samples used in this study were products purchased from manufacturers. Tracing back to the raw materials may be effective to gain more insight into the elemental content trends in herbal medicines.

Conclusion

In this study, we performed EDXRF analysis of herbal medicines to determine their quantitative ability compared with ICP-MS. Quantitative analyses of heavy metals in various herbal medicines derived from plants, animals, and minerals were performed using EDXRF and ICP-MS. Although the results were in good agreement, the quantitative EDXRF values of the minerals tended to be slightly less accurate than those of plants. EDXRF enables rapid and easy determination of heavy metals without troublesome pretreatments in herbal medicines. Furthermore, EDXRF analysis is expected to improve quality control and evaluation when combined with simultaneous multi-elemental analysis. Enrichment of EDXRF data using more herbal medicine samples in future studies will elucidate the utility of EDXRF for heavy metal analysis.

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Author contributions Ryoko Arai acquired EDXRF and ICP-MS data and wrote the manuscript. Hiroyuki Kamakura acquired ICP-MS data. Naoko Masumoto coordinated the study and interpreted the results. Michiho Ito supervised the research and reviewed the manuscript. All authors critically revised the manuscript, commented on drafts of the manuscript, and approved the final report.

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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